Veronica Boyce

vboyce@mit.edu or 650-933-0841
vboyce.github.io

EDUCATION

Stanford University **Stanford, CA**
Doctor of Philosophy in Psychology September 2025
Dissertation: *Constraints on the emergence and interpretability of linguistic conventions*
Massachusetts Institute of Technology **Cambridge, MA**
Bachelor of Science in Brain and Cognitive Sciences June 2019

EMPLOYMENT

MIT, Computational Psycholinguistics Lab **Cambridge, MA**
NSF SBE Postdoctoral Research Fellow Jan 2026 – present
Stanford University, Language and Cognition Lab **Stanford, CA**
Postdoctoral Researcher Sep 2025 – Jan 2026
MIT, Computational Psycholinguistics Lab **Cambridge, MA**
Research Support Associate June 2019 – July 2020

AWARDS, FELLOWSHIPS, & GRANTS

NSF SBE Postdoctoral Research Fellowship Jan 2026 – Jan 2028
Open Science Innovator Prize, *Stanford Center for Open and Reproducible Science* 2025
The Albert H. and Barbara R. Hastorf Prize for Teaching, *Stanford Department of Psychology* 2025

PUBLICATIONS

V. Boyce, R. Z. Sparks, and M. C. Frank (2026). "Preschoolers can coordinate with each other to communicate about novel referents". *Open Mind*. doi.org/10.1162/opmi.a.399.

M. C. Frank, V. A. Marchman, C. A. Bergey, **V. Boyce**, M. Braginsky, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, N. Ram, R. Z. Sparks, A. Steffan, A. W. M. Tan, and M. Zettersten (2026). "Continuous developmental changes in word recognition support language learning across early childhood". *eLife*. elifesciences.org/articles/109636.

A. N. Meltzoff, S. Farwaha, V. Slaughter, E. A. Simpson, R. Brooks, M. Nielsen, H. A. Baumgartner, L. Bayet, S. Benavides-Varela, **V. Boyce**, K. Byers-Heinlein, J. Dönicke, B. P. Flaherty, S. H. Forbes, M. C. Frank, A. Geraci, K. Hamlin, M. Heimann, N. Keven, J. Lany, J. R. Lelonkiewicz, T. S. Leung, S. Maylott, D. Moreau, E. Quadrelli, P. J. Reschke, C. Turati, S. Vincini, S. M. Weiss, T. S. Zamuner, and G. Zeng (2026). "ManyBabies 6: A Multi-Lab Investigation of Neonatal and Early Imitation". *Developmental Science*. Stage 1 Registered Report, in-principle acceptance, preprint, osf.io/preprints/psyarxiv/fe7pj_v1.

M. Soderstrom, J. Rocha-Hidalgo, L. E. Muñoz, A. Bochynska, J. F. Werker, B. Skarabela, A. Seidl, Y. Ryjova, J. L. Rennels, C. E. Potter, M. Paulus, M. Ota, N. M. Olesen, K. M. Nave, J. Mayor, A. Martin, L. C. Machon, C. Lew-Williams, E.-S. Ko, H. Kim, N. Kartushina, M. Kammermeier, A. Jessop, J. F. Hay, N. Havron, E. E. Hannon, J. K. Hamlin, N. Gonzalez-Gomez, A. Gampe, T. Fritzsche, M. C. Frank, S. Durrant, C. Davies, C. Cashon, K. Byers-Heinlein, **V. Boyce**, A. K. Black, C. Bergmann, L. Anderson, M. K. Alshakhori, A. H. Al-Hoorie, and A. S. M. Tsui (2025). "Testing the relationship between preferences for infant-directed speech and vocabulary development: A multi-lab study". *Journal of Child Language*. doi.org/10.1017/S0305000924000254.

V. Boyce, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2024). "Interaction Structure Constrains the Emergence of Conventions in Group Communication". *PNAS*. doi.org/10.1073/pnas.2403888121.

V. Boyce, B. Prystawski, A. B. Abutto, E. M. Chen, Z. Chen, H. Chiu, I. Ergin, A. Gupta, C. Hu, B. Kemmann, N. Klevak, V. Y. Q. Lua, M. Mazzaferro, K. Mon, D. Ogunbamowo, A. Pereira, J. Troutman, S. Tung, R. Uricher, and M. C. Frank (2024). "Estimating the replicability of psychology experiments after an initial failure to replicate". *Collabra: Psychology*. doi.org/10.1525/collabra.125685.

A. Chuey, **V. Boyce**, A. Cao, and M. C. Frank (2024). "Conducting Developmental Research Online vs. in-Person: A Meta-Analysis". *Open Mind*. doi.org/10.1162/opmi_a_00147.

V. Boyce and R. P. Levy (2023). "A-Maze of Natural Stories: Comprehension and Surprisal in the Maze Task". *Glossa Psycholinguistics*. escholarship.org/uc/item/6vh9d8zm.

V. Boyce, M. Mathur, and M. C. Frank (2023). "Eleven Years of Student Replication Projects Provide Evidence on the Correlates of Replicability in Psychology". *Royal Society Open Science*. doi.org/10.1098/rsos.231240.

M. Zettersten, D. Yurovsky, T. L. Xu, S. Uner, A. S. M. Tsui, R. M. Schneider, A. N. Saleh, S. C. Meylan, V. A. Marchman, J. Mankewitz, K. MacDonald, B. Long, M. Lewis, G. Kachergis, K. Handa, B. deMayo, A. Carstensen, M. Braginsky, **V. Boyce**, N. S. Bhatt, C. A. Bergey, and M. C. Frank (2023). "Peekbank: An Open, Large-Scale Repository for Developmental Eye-Tracking Data of Children's Word Recognition". *Behavior Research Methods*. doi.org/10.3758/s13428-022-01906-4.

V. Boyce, R. Futrell, and R. P. Levy (2020). "Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty". *Journal of Memory and Language*. doi.org/10.1016/j.jml.2019.104082.

CONFERENCE PROCEEDINGS

A. W. M. Tan, B. Prystawski, and **V. Boyce** (2026). "Which one is banana man? Evaluating vision-language models in multi-turn pragmatic interpretation". *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. Talk, aclanthology.org/2026.emnlp-main.548.

V. Boyce, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). "Idiosyncratic but not opaque: Linguistic conventions formed in reference games are interpretable by naïve humans and vision–language models". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/16c4n85d.

V. Boyce, R. Z. Sparks, Y. Mofor, and M. C. Frank (2025). "Preschoolers can form conventional pacts with each other to communicate about novel referents". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, osf.io/dpgfw_v1.

A. W. M. Tan, B. Prystawski, **V. Boyce**, and M. C. Frank (2025). "Context informs pragmatic interpretation in vision-language models". *CogInterp workshop NeurIPS 2025*. Poster, preprint at arxiv.org/abs/2511.03908.

V. Boyce and M. C. Frank (2023). "Communicative Reduction in Referring Expressions within a Multi-Player Negotiation Game". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/8dq8c2s6.

V. Boyce, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2022). "Two's Company but Six Is a Crowd: Emergence of Conventions in Multiparty Communication Games". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Talk, osf.io/preprints/psyarxiv/xd7ep.

J. Mankewitz, **V. Boyce**, B. Waldon, G. Loukatou, D. Yu, J. Mu, N. D. Goodman, and M. C. Frank (2021). "Multi-Party Referential Communication in Complex Strategic Games". *Meaning in Context: Pragmatic Communication in Humans and Machines workshop. NeurIPS.* PsyArXiv. Talk, osf.io/preprints/psyarxiv/tfb3d.

M. Zettersten, C. A. Bergey, N. S. Bhatt, **V. Boyce**, M. Braginsky, A. Carstensen, B. deMayo, G. Kachergis, M. Lewis, B. Long, K. MacDonald, J. Mankewitz, S. C. Meylan, A. N. Saleh, R. M. Schneider, A. S. M. Tsui, S. Uner, T. L. Xu, D. Yurovsky, and M. C. Frank (2021). "Peekbank: Exploring Children's Word Recognition through an Open, Large-Scale Repository for Developmental Eye-Tracking Data". *Proceedings of the Annual Meeting of the Cognitive Science Society*. Poster, escholarship.org/uc/item/9j05h4n1.

V. Boyce, A. Aravind, and M. Hackl (2017). "Lexical and Syntactic Effects on Auxiliary Selection: Evidence from Child French". *Boston University Conference on Language Development*. Talk, lingref.com/bucld/41/BUCLD41-08.pdf.

PREPRINTS (UNDER REVIEW)

V. Boyce, V. A. Marchman, H. A. Baumgartner, C. A. Bergey, M. Braginsky, F. Giovannetti, G. Kachergis, J. Mankewitz, S. C. Meylan, B. Prystawski, R. Z. Sparks, A. Steffan, A. W. M. Tan, M. C. Frank, and M. Zettersten (2026). "Data-driven recommendations for increasing reliability and validity in measures of infant word recognition". Preprint, under review, osf.io/preprints/psyarxiv/28gq4_v2.

V. Boyce, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, M. C. Frank, and R. D. Hawkins (2026). "Refbank: an open repository of iterated reference game data". Preprint, under review, osf.io/preprints/psyarxiv/d93gy_v1.

V. Boyce, A. Acharya, and M. C. Frank (2025). "Use of communication channels in 2x2 strategic games". White paper, osf.io/preprints/psyarxiv/y6ahw_v1.

L. Röseler, L. Wallrich, H. Hartmann, L. Altegoer, **V. Boyce**, S. M. Field, J. Goltermann, J. Hüffmeier, C. R. Pennington, M.-M. Pittelkow, P. Silverstein, D. van Ravenzwaaij, and F. Azevedo (2025). "Handbook on Conducting Reproduction and Replication Studies". Preprint, under revision following MetaROR peer review, doi.org/10.5281/zenodo.16990114.

CONFERENCE PRESENTATIONS

V. Boyce, A. W. M. Tan, J. Mankewitz, F. Feng, B. Fernandez, R. Fried, B. Prystawski, M. C. Frank, and R. D. Hawkins (2026). "Refbank: an open repository of iterated reference games". The 39th Annual Conference on Human Sentence Processing. Poster.

V. Boyce and R. P. Levy (2025). "Linguistic, but not visual, discourse context modulates syntactic parsing in the Maze". The 38th Annual Conference on Human Sentence Processing. Poster.

V. Boyce, B. Prystawski, A. W. M. Tan, and M. C. Frank (2025). "Context matters: History and informativity affect understanding of opaque references". The 38th Annual Conference on Human Sentence Processing. Poster.

J. Vigly, **V. Boyce**, M. Socolof, J. Michaelov, T. von der Malsburg, and R. P. Levy (2025). "Implicit gender bias in linguistic descriptions for expected events: The case of the 2024 United States presidential election". The 38th Annual Conference on Human Sentence Processing. Poster.

V. Boyce, I. Chen, R. Z. Sparks, M. Perez, and M. C. Frank (2024). "4-5 year old children can successfully communicate using ad-hoc referential expressions". CAMP 6. Talk.

V. Boyce, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). "Thick Feedback Facilitates Referential Coordination in Larger Groups". CAMP 5. Talk.

V. Boyce, R. D. Hawkins, N. D. Goodman, and M. C. Frank (2023). "Thick Feedback Facilitates Referential Coordination in Larger Groups". HSP. Poster.

T. Poppels, **V. Boyce**, C. Ajunwa, T. von der Malsburg, and R. P. Levy (2021). "Bias against 'she' Pronouns Can Be Rapidly Overcome by Changing Event Expectations". CUNY Human Sentence Processing Conference. Short talk.

V. Boyce and R. P. Levy (2020). "A-Maze of Natural Stories: Texts Are Comprehensible Using the Maze Task". Architectures and Mechanisms for Language Processing. Talk.

V. Boyce, R. Futrell, and R. P. Levy (2019). "Maze Made Easy: Better and Easier Measurement of Incremental Processing Difficulty". CUNY Human Sentence Processing Conference. Poster.

V. Boyce, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). "Female Gender Is Consistently Under-Expressed in Pronoun Production and Under-Inferred in Comprehension". Linguistic Society of America Annual Meeting. Poster.

V. Boyce, T. von der Malsburg, T. Poppels, and R. P. Levy (2019). "Remember 'Him', Forget 'Her': Gender Bias in the Comprehension of Pronominal Referents." CUNY Human Sentence Processing Conference. Poster.

V. Boyce, T. von der Malsburg, T. Poppels, and R. P. Levy (2018). "Implicit Gender Biases in the Production and Comprehension of Pronominal References". CUNY Human Sentence Processing Conference. Poster.

INVITED TALKS

Harvard LangCog Talk Series, Harvard University	Sep 2026
Goldenberg Lab, Harvard Business School	Nov 2025
Language and Cognitive Development Lab, UC Berkeley	Oct 2025
Saxelab, MIT	Feb 2023
QuantLang Lab, UC Irvine	April 2021

TEACHING

Stanford, Department of Psychology **Stanford, CA**

TA for PSYCH 125: Research Methods in Psychology Jan 2024 – March 2024
First offering of course; designed and led course replication projects with bi-weekly milestones

TA for PSYCH 251: Experimental Methods Sep 2023 – Dec 2023
Designed and supervised re-replication option for course project
Led collective writing of joint paper reporting findings across projects (see Publications)

TA for PSYCH 251: Experimental Methods Sep 2022 – Dec 2022

Guest Lecturer, PSYCH 251: Experimental Methods 2023, 2025
Topic: meta-analysis and meta-science

TA for PSYCH 45: Introduction to Learning and Memory April 2023 – June 2023

TA for PSYCH 220B: Probabilistic Models of Cognition: Language Sep 2021 – Dec 2021

MIT, Department of Brain and Cognitive Sciences **Cambridge, MA**

TA for Computational Psycholinguistics February 2019 – May 2019

MENTORING

Benji Fernandez (Stanford, undergraduate honors thesis in progress, co-advisor) 2025 – 2026
Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints

Fiona Feng (Queens College undergraduate research assistant) 2025 – 2026
Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints

Ryder Fried (Stanford undergraduate research assistant) 2025
Co-author, poster at HSP and manuscript, see Conference Presentations & Preprints

Spurti Nimbali (Stanford undergraduate research assistant) 2025

Zihan Wang (Stanford undergraduate research assistant) 2025

Yannick Mofor (Stanford undergraduate research assistant) 2024
Co-author, CogSci proceedings, see Conference Proceedings

Rebecca Pizzitola (Stanford undergraduate research assistant) 2024

Ilaria Chen (Stanford undergraduate research assistant) 2023
Co-author, talk at CAMP, see Conference Presentations

Malia Perez (Stanford undergraduate research assistant) 2023
Co-author, talk at CAMP, see Conference Presentations
Now PhD student at UNC Chapel Hill

Brittany Linus (Stanford undergraduate research assistant) 2022

Chelsea Ajunwa (MIT undergraduate research assistant) 2020
Co-author, short talk at HSP, see Conference Presentations
Now PhD student at Northeastern University

Arun Wongprommoon (MIT undergraduate research assistant) 2019

AD HOC REVIEWING

Journals: Replication Research (2026); Journal of Personality and Social Psychology: Attitudes and Social Cognition (2026); Open Mind (2025–2026); Infant Behavior and Development; Language, Cognition, and Neuroscience (2024); Royal Society Open Science (2024); Journal of Experimental Psychology: Learning, Memory, and Cognition (2023); Journal of Memory and Language (2023);

Glossa Psycholinguistics (2021)
Co-reviewed (with Michael C. Frank): Psychological Science (2025), Perspectives on Psychological Science (2025), Nature Human Behaviour (2024), Glossa Psycholinguistics (2023)
Conferences: CogSci (2021–2026), HSP (2026), California Meeting of Psycholinguists (2023), Society for Computation in Linguistics (2020–2026), Association for Computational Linguistics (2019)

REFERENCES

Dr. Roger Levy, Department of Brain and Cognitive Sciences, MIT
rplevy@mit.edu, (617) 253-5763

Dr. Michael C. Frank, Department of Psychology, Stanford University
mcfrank@stanford.edu, (650) 724-4003

Dr. Robert Hawkins, Department of Linguistics, Stanford University
rdhawkins@stanford.edu, (217) 549-6923